

Information retrieval is the activity of obtaining information system resources that are relevant to an information need from a collection of those resources. Searches can be based on full-text or other content-based indexing. Automated information retrieval systems are used to reduce what has been called information overload. Many universities and public libraries use retrieval systems to provide access to books, journals, and other documents.

An information retrieval process begins when a user enters a query into the system. Queries are formal statements of information needs, for example search strings in web search engines. In information retrieval a query does not uniquely identify a single object in the collection. Instead, several objects may match the query, perhaps with different degrees of relevance.

An object is an entity that is represented by information in a content collection or database. User queries are matched against the database information. Depending on the application the data objects may be, for example, text documents, images, audio, mind maps or videos. Often the documents themselves are not kept or stored directly in the IR system, but are instead represented in the system by document surrogates or metadata.

Most IR systems compute a numeric score on how well each object in the database matches the query, and rank the objects according to this value. The top ranking objects are then shown to the user. The process may then be iterated if the user wishes to refine the query. The effectiveness of a retrieval system is commonly measured with precision, recall, and mean average precision over a set of queries with known relevance judgments.

Boolean retrieval was the dominant model in early systems. A Boolean query returns the set of documents satisfying a logical expression over terms, with no notion of ranking. This works well for expert users who understand both the collection and the query language, but it produces either far too many or far too few results for most people, which is why ranked retrieval replaced it in general-purpose search.